

Olist E-Commerce Analysis

Complete Project Documentation — from Raw Data to Dashboard

Power Query • Data Modeling • Power BI Dashboard • Insights

Project Overview

This document covers the full project end-to-end: cleaning the 9 raw Olist datasets in Power Query, building the relational data model, constructing a 6-page interactive Power BI dashboard, and the business findings derived from it.

R\$16.0M Total Revenue	99,441 Total Orders	R\$154 Avg Order Value	4.09 / 5 Avg Review Score
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Data Sources

The project uses the Olist Brazilian E-Commerce public dataset, composed of 9 related CSV files.

olist_customers	Customer IDs, unique customer IDs, city, state, zip code prefix (~99,441 rows)
olist_geolocation	Zip code prefixes with latitude/longitude and city/state (~1,000,163 rows before cleanup)
olist_order_items	Line items per order: product, seller, price, freight value (~112,650 rows)
olist_order_payments	Payment type, installments, and value per order (~103,886 rows)
olist_order_reviews	Review score and dates per order (~104,719 rows)
olist_orders	Order status and full timestamp lifecycle (~99,441 rows) — the central fact table
olist_products	Product category (Portuguese), dimensions, weight, photo count (~32,951 rows)
olist_sellers	Seller ID, city, state, zip code prefix (~3,095 rows)
product_category_name_translation	Portuguese-to-English category name mapping (71 rows)

Data Preparation — Power Query

Every table was cleaned and typed correctly in Power Query before being loaded into the model.

orders (central fact table)

- Converted all date columns to **Date/Time** : order_purchase_timestamp, order_approved_at, order_delivered_carrier_date, order_delivered_customer_date, order_estimated_delivery_date
- Cleaned order_status with Trim and Clean
- Added a calculated column **Delivery Delay** = order_delivered_customer_date – order_estimated_delivery_date, typed as **Duration**

customers

- Set customer_id, customer_unique_id, customer_city, customer_state to **Text**
- Set customer_zip_code_prefix to **Text** (not Number, to preserve leading zeros)
- Cleaned customer_city with Trim and Clean
- Built a separate reference table ("Number of Orders") grouping by customer_unique_id to count orders per real customer

geolocation

- Set geolocation_zip_code_prefix to **Text** ; lat/lng to **Decimal Number**
- Grouped by zip code prefix, reducing ~1 million rows to ~19,000, averaging latitude and longitude (Max used for city/state)

order_items

- Set order_id, product_id, seller_id to **Text** ; order_item_id to **Whole Number**
- Set shipping_limit_date to **Date/Time** ; price and freight_value to **Decimal Number**

order_payments

- Set order_id, payment_type to **Text** ; payment_sequential, payment_installments to **Whole Number** ; payment_value to **Decimal Number**
- Cleaned payment_type with Trim and Clean

order_reviews

- Set review_id, order_id to **Text** ; review_score to **Whole Number** (verified range 1–5)
- Set review_creation_date, review_answer_timestamp to **Date/Time**
- Removed review_comment_title and review_comment_message (mostly empty, not used for text analysis)

products

- Set dimension/measurement columns to **Whole Number**
- Merged with product_category_name_translation on product_category_name (Left Outer join)
- Expanded the merged column, keeping only product_category_name_english
- Removed the original Portuguese category column after extracting the English version

sellers

- Set seller_id, seller_city, seller_state to **Text** ; seller_zip_code_prefix to **Text**
- Cleaned seller_city with Trim, Clean, then Capitalize Each Word

product_category_name_translation

- Fixed a hidden BOM character that broke the header row (columns showed as Column1/Column2)
- Reapplied "Use First Row as Headers" to restore product_category_name and product_category_name_english



customer_zip_code_prefix → geolocation_zip_code_prefix (many-to-one)

customers →
geolocation

Additional calculated columns (DAX)

Two calculated columns were added directly to the **orders** table in Data View to support the Delivery Performance page:

```
Delivery Status =  
IF(  
    ISBLANK(olist_orders[Delivery Delay]),  
    "Not Delivered",  
    IF(olist_orders[Delivery Delay] < 0, "On Time", "Late")  
)
```

```
Delivery Days = DATEDIFF(olist_orders[order_purchase_timestamp],  
    olist_orders[order_delivered_customer_date], DAY)
```

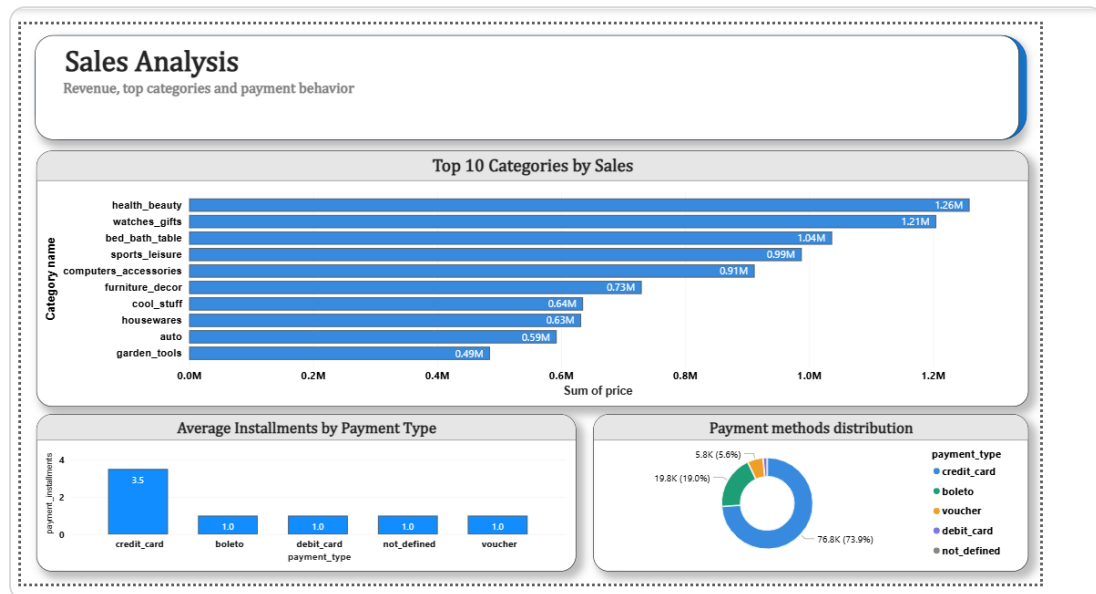
Dashboard — 1. Overview



KPI cards, revenue trend, top 5 states, and order status breakdown.

- 4 KPI cards with icons: Total Revenue (), Total Orders (), Average Order Value (), Average Review Score ()
- Line chart of revenue by Year + Month (drilled down from the default Year/Quarter/Month/Day hierarchy)
- Horizontal bar chart: Top 5 selling states by payment value
- Donut chart: order status distribution, colored with delivered in green as the dominant category
- Consistent page background (#F2F2F0) and white card backgrounds established here and reused on all later pages

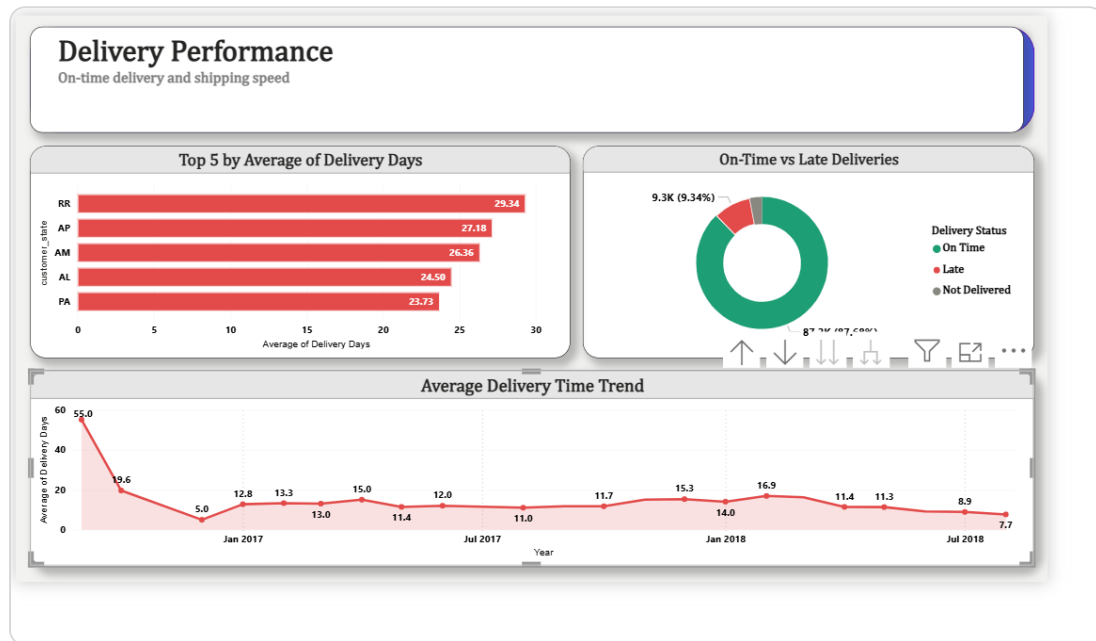
Dashboard — 2. Sales Analysis



Top 10 categories, average installments by payment type, and payment method distribution.

- Top 10 Categories by Sales — horizontal bar chart using price (Sum) from order_items joined through products, filtered with Top N = 10
- Average Installments by Payment Type — column chart, colored to match the donut chart's payment type colors
- Payment Methods Distribution — donut chart showing credit_card as the dominant method (73.9%)

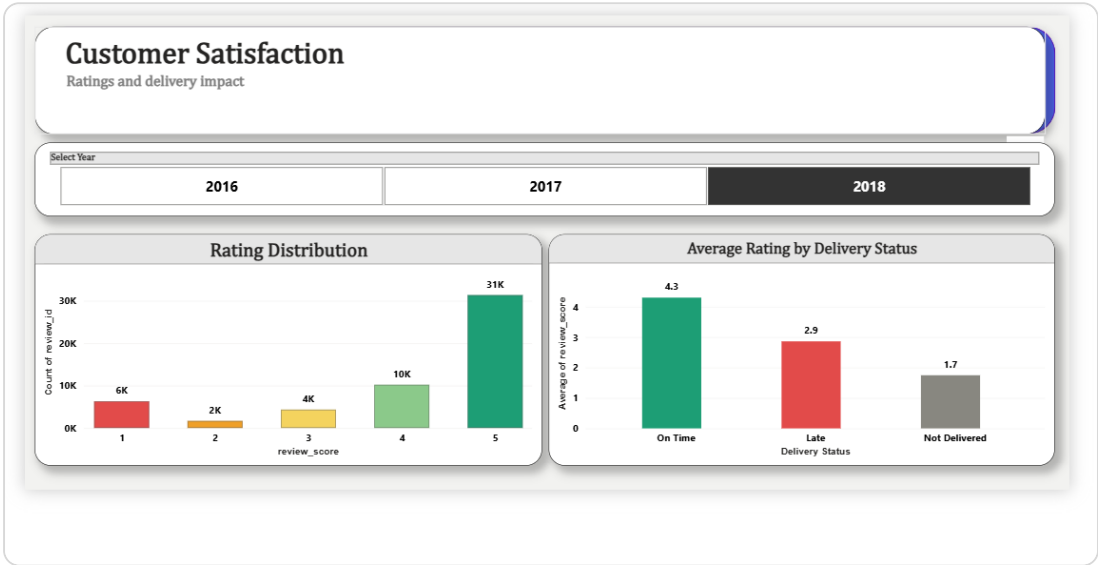
Dashboard — 3. Delivery Performance



Slowest states, on-time vs late split, and the delivery time trend.

- Top 5 Slowest States — bar chart of Average Delivery Days by customer_state, colored solid red to signal a problem area
- On-Time vs Late Deliveries — donut chart built from the Delivery Status calculated column
- Average Delivery Time Trend — line chart showing delivery days improving from 55 days at launch (2016) down to under 10 days by late 2018

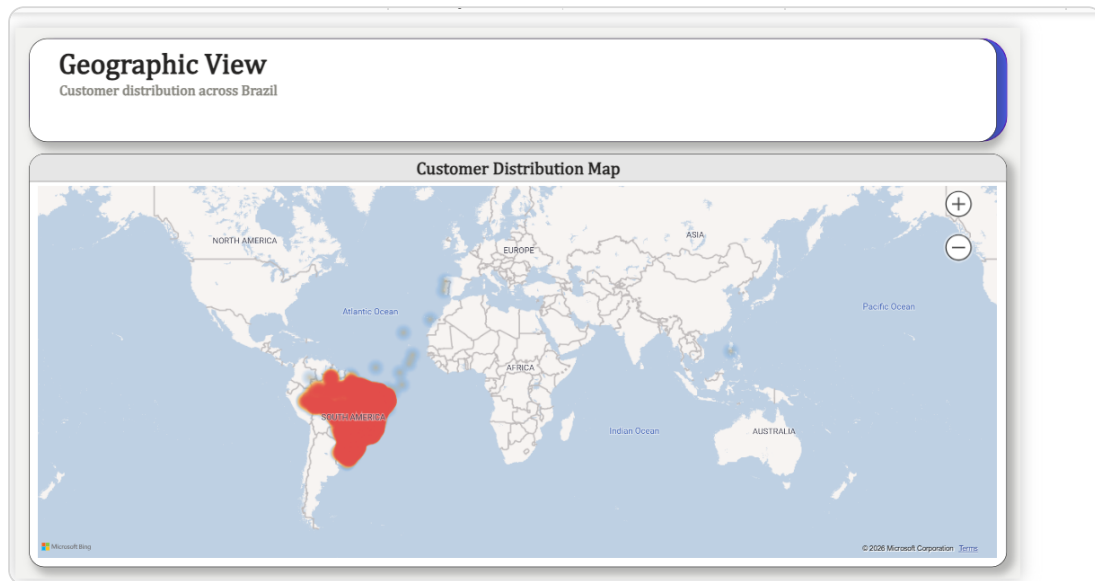
Dashboard — 4. Customer Satisfaction



Rating distribution, delivery impact on ratings, and a year slicer.

- Year slicer (Tile style, single-select) at the top of the page, filtering both visuals below
- Rating Distribution — column chart of review_score counts, colored on a red-to-green gradient (1 star red, 5 stars green)
- Average Rating by Delivery Status — column chart confirming that late deliveries significantly reduce satisfaction (4.3 On Time vs 2.8 Late)

Dashboard — 5. Geographic View



Heat map of customer concentration, built from the cleaned geolocation table.

- Heat map visual using Average Lat / Average Lng from the geolocation table (set as Latitude/ Longitude data categories)
- Color gradient from blue (low density) through yellow to red (high density)
- Confirms overwhelming customer concentration within Brazil, consistent with the dataset's scope

Key Findings & Recommendations

1 Delivery & Logistics — Top Priority

Finding: Late deliveries drop the average customer rating from 4.3 to 2.8 stars — a 1.5-point decline.

Recommendation: Invest in improving delivery logistics, especially in the slowest states (RR, AP, AM, AL, PA), which average 24–29 days versus 7–15 days elsewhere.

2 Geographic Concentration — Growth Opportunity

Finding: São Paulo (SP) alone generates R\$6.0M in sales — over 3x the next closest state (RJ at R\$2.1M).

Recommendation: Explore targeted marketing campaigns in underrepresented states (RS, PR) to reduce over-reliance on a single region.

3 Top-Selling Categories — Resource Allocation

Finding: health_beauty, watches_gifts, and bed_bath_table are the top 3 categories by sales (R\$1.26M, R\$1.21M, R\$1.04M respectively).

Recommendation: Prioritize inventory availability and negotiate better supplier terms for these categories specifically.

4 Payment Behavior — Checkout Experience

Finding: Credit card accounts for 73.9% of payments with an average of 3.5 installments, while other methods average just 1 installment.

Recommendation: Expand installment options for other payment methods (e.g., boleto) to encourage larger purchases.

5 Data Quality Note — Technical

Finding: Data for the last two months (Sep/Oct 2018) appears incomplete, causing a misleading sharp drop in the revenue trend chart.

Recommendation: Verify data completeness before drawing conclusions based on the most recent period in any report.